Math 110 — Week 4

Isomorphisms, inverses, products, quotient spaces (Notes Ch. 7–8, LADR 3D–3E, HW4)

§1 Facts used from Weeks 1–3

(W1) A sum $U_1 + \cdots + U_m$ is direct exactly when $u_1 + \cdots + u_m = 0$ with $u_i \in U_i$ forces every $u_i = 0$; then $\dim(U_1 \oplus \cdots \oplus U_m) = \sum \dim U_i$ (Weeks 1–2).

(W2) An independent list extends to a basis. In a space of dimension n, n independent vectors form a basis, and so do n spanning vectors (Week 2).

(W3) Given a basis $v_1, \ldots, v_n$ of V and any $w_1, \ldots, w_n \in W$, there is exactly one linear T with $Tv_j = w_j$. Two linear maps that agree on a basis are equal (Week 3, Prop. 21).

(W4) T is injective $\iff \operatorname{null} T = \{0\}$ (Week 3, Prop. 23).

(W5) Rank–nullity: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ (Week 3, Prop. 24).

(W6) If $\dim V = \dim W < \infty$: T injective $\iff$ T surjective (Week 3, Prop. 25).

§2 This week

§2.1 Isomorphisms and inverses (Ch. 7.1)

The question. When are two vector spaces "the same" as far as linear algebra is concerned, even though their elements look different? $\mathcal{P}_2(\mathbb{R})$ and $\mathbb{R}^3$: the polynomial $a + bx + cx^2$ and the triple (a, b, c) carry the same information, and adding or scaling on one side matches adding or scaling on the other.

Definition (Def. 21 / LADR 3.59, 3.69)**.** A linear map $T\colon V \to W$ is an **isomorphism** (is **invertible**) if there is a linear map $S\colon W \to V$ with $TS = \mathrm{id}_W$ and $ST = \mathrm{id}_V$. Then V and W are **isomorphic**, written $V \cong W$.

Proposition (Prop. 28 / LADR 3.60: the inverse is unique)
If T is an isomorphism, the map S is unique. It is written T^{-1}.

Proof. If S_1, S_2 both work, then $S_1 = S_1(TS_2) = (S_1T)S_2 = S_2$. □

- The computation uses only $S_1T = \mathrm{id}$ and $TS_2 = \mathrm{id}$. So: *if T has a left inverse and a right inverse, they are equal.*
- With only one side, uniqueness fails. $A = \begin{pmatrix} 1 & 2 \end{pmatrix} : \mathbb{F}^2 \to \mathbb{F}$ has no left inverse (it is not injective) and infinitely many right inverses:

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 1 - 2x \\ x \end{pmatrix} = 1$$

for every x.

Proposition (Prop. 29 / LADR 3.63: isomorphism = linear bijection)
A linear map $T\colon V \to W$ is an isomorphism if and only if it is injective and surjective.

The idea

A bijection always has an inverse *function*. The definition asks for a *linear* inverse, so the content is that the inverse function of a linear bijection is automatically linear.

How do you prove $T^{-1}(\lambda w_1 + w_2)$ equals $\lambda T^{-1}(w_1) + T^{-1}(w_2)$ when you have no formula for T^{-1}? Both sides are vectors in V, and T is injective, so it is enough to show T sends them to the same vector. Applying T removes every T^{-1}. This move (to compare two vectors, apply an injective map to both) is used again and again.

Proof. 1. ($\Rightarrow$) If $Tv = 0$ then $v = T^{-1}(Tv) = T^{-1}(0) = 0$: injective. Any w equals $T(T^{-1}w)$: surjective.

2. ($\Leftarrow$) For $w \in W$ let $T^{-1}(w)$ be the unique v with $Tv = w$. Then $T(T^{-1}w) = w$ and $T^{-1}(Tv) = v$.

3. T^{-1} is linear:

$$T\big(T^{-1}(\lambda w_1 + w_2)\big) = \lambda w_1 + w_2 = \lambda T(T^{-1}w_1) + T(T^{-1}w_2) = T\big(\lambda T^{-1}w_1 + T^{-1}w_2\big),$$

and T is injective.

□

Proposition (Prop. 30 / LADR 3.70: dimension classifies finite-dimensional spaces)
Finite-dimensional V and W are isomorphic if and only if $\dim V = \dim W$.

The idea

A basis $v_1, \ldots, v_n$ of V gives every vector a unique coordinate column, and that is an isomorphism $V \to \mathbb{F}^n$. So every n-dimensional space is a copy of $\mathbb{F}^n$, and two n-dimensional spaces are copies of each other. To build the isomorphism directly, send a basis of V to a basis of W; the inverse sends the basis of W back.

Proof. 1. ($\Rightarrow$) If T is an isomorphism, $\operatorname{null} T = \{0\}$ and $\operatorname{range} T = W$, so by (W5) $\dim V = 0 + \dim W$.

2. ($\Leftarrow$) Choose bases $v_1, \ldots, v_n$ of V and $w_1, \ldots, w_n$ of W. By (W3) there are linear maps T with $Tv_j = w_j$ and S with $Sw_j = v_j$.

3. $STv_j = v_j$ and $TSw_j = w_j$. So ST and id_V agree on a basis, and TS and id_W agree on a basis. Hence $ST = \mathrm{id}_V$, $TS = \mathrm{id}_W$ (W3).

□

Proposition (Prop. 31 / LADR 3.65: with equal dimensions, one condition is enough)
If $\dim V = \dim W < \infty$ and $T \in \mathcal{L}(V, W)$, the following are equivalent: T is an isomorphism; T is injective; T is surjective.

Proof. Injective and surjective are equivalent here (W6), and both together mean isomorphism (Prop. 29). □

Example (Ex. 34: interpolation)
Given points $(x_0, y_0), \ldots, (x_n, y_n)$ with distinct x_j, there is exactly one $p \in \mathcal{P}_n(\mathbb{R})$ with $p(x_j) = y_j$ for all j.

1. Let $T\colon \mathcal{P}_n(\mathbb{R}) \to \mathbb{R}^{n+1}$, $T(p) = (p(x_0), \ldots, p(x_n))$. Both spaces have dimension $n+1$.
2. If $Tp = 0$, then p has the $n+1$ distinct zeros $x_0, \ldots, x_n$, so $(x - x_0)\cdots(x - x_n)$ divides p. A nonzero p would then have degree $\geq n+1$. So $p = 0$ and T is injective.
3. By Prop. 31, T is bijective: every $(y_0, \ldots, y_n)$ is $T(p)$ for exactly one p.

Example (Ex. 35 / LADR 3.67: existence from injectivity)

For every polynomial q there is a polynomial p with $\big((x^2+5x+7)p\big)'' = q$.

1. Choose m with $q \in \mathcal{P}_m(\mathbb{R})$ and let $T\colon \mathcal{P}_m(\mathbb{R}) \to \mathcal{P}_m(\mathbb{R})$, $T(p) = \big((x^2+5x+7)p\big)''$. Multiplying raises the degree by 2 and differentiating twice lowers it by 2, so T maps $\mathcal{P}_m(\mathbb{R})$ to itself.
2. If $Tp = 0$, then $(x^2+5x+7)p$ has degree ≤ 1. For $p \neq 0$ its degree is ≥ 2. So $p = 0$: T is injective.
3. T is an operator on a finite-dimensional space, so it is surjective. No coefficients were solved for.

Ex. 36: in infinite dimensions this fails. Differentiation $\mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is surjective and not injective.

The 2×2 determinant (computational aside). For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ define $\det A = ad - bc$. (This is only a formula for now; determinants in general come at the end of the course.)

- Direct multiplication:
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad-bc & -ab+ba \\ cd-dc & -cb+da \end{pmatrix} = (ad-bc)I,$$
and the same in the other order.
- If $\det A \neq 0$: $A^{-1} = \frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$. Swap the diagonal entries, negate the other two, divide by $\det A$.
- If $\det A = 0$, then A is not invertible. If $(a,b) \neq (0,0)$:
$$A\begin{pmatrix} b \\ -a \end{pmatrix} = \begin{pmatrix} ab-ba \\ cb-da \end{pmatrix} = 0,$$
so A is not injective. If $a = b = 0$: every output has first coordinate 0, so A is not surjective.
- Hence A is invertible $\iff \det A \neq 0$.

§2.2 One-sided inverses (Ch. 7.3)

The question. If ST is invertible, must S and T be? If $ST = I$, is $TS = I$?

Proposition (Prop. 32 / LADR 3D: inverses and products)

Let $S, T \in \mathcal{L}(V)$.

(i) If S and T are invertible, so is ST, and $(ST)^{-1} = T^{-1}S^{-1}$.

(ii) If ST is invertible, then S is surjective and T is injective. If V is finite-dimensional, then S, T and TS are all invertible.

The idea

In ST, T acts first and S acts last. If T sent two vectors to the same place, so would ST; so ST injective forces T injective. Every output of ST is an output of S; so ST surjective forces S surjective. Neither argument says anything about the other map, and in infinite dimensions the other half can fail (Ex. 37, 38). In finite dimensions, Prop. 31 upgrades each half to invertibility.

Proof. 1. (i)
$$(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}T = I$$
and $(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = I$.

2. (ii) Any v is
$$v = (ST)(ST)^{-1}v = S\big(T(ST)^{-1}v\big) \in \operatorname{range} S.$$

3. If $Tv = 0$, then $v = (ST)^{-1}(ST)v = (ST)^{-1}S(0) = 0$.

4. If $\dim V < \infty$, Prop. 31 makes S and T invertible, and (i) makes TS invertible.

□

Proposition (Prop. 33 / LADR 3D.11: one side is enough in finite dimensions)
If V is finite-dimensional and $S, T \in \mathcal{L}(V)$, then $ST = I \iff TS = I$.

Proof. If $ST = I$, then ST is invertible, so T is invertible (Prop. 32). Then $S = S(TT^{-1}) = (ST)T^{-1} = T^{-1}$, so $TS = TT^{-1} = I$. The converse is the same with the roles exchanged. □

- Ex. 38: on $\mathcal{P}(\mathbb{R})$ let $S(p) = p'$ and $T(p)(x) = \int_0^x p(t)\,dt$. Then $ST = I$, but $TS(p) = p - p(0) \neq p$ in general.
- Ex. 37: on $\mathcal{P}(\mathbb{R})$ let $S(p) = p'$ and $T(p)(x) = \int_0^{x+1} p(t)\,dt$. Then $ST(p)(x) = p(x+1)$, which is invertible (inverse $p(x) \mapsto p(x-1)$), yet S is not injective, T is not surjective ($\operatorname{range} T = \{p : p(-1) = 0\}$), and TS sends constants to 0.
- Even then, T is an isomorphism from V onto $\operatorname{range} T$, and $(ST)^{-1} = T^{-1}(S|_{\operatorname{range} T})^{-1}$.

§2.3 Products (Ch. 8.1)

The question. Given unrelated vector spaces $V_1, \dots, V_m$, how do we build one space that contains a copy of each, side by side? ($\mathbb{R}^n$ is built from n copies of $\mathbb{R}$ this way.)

Definition (Prop. 34 / LADR 3.87, 3.89)**.** The **product** $V_1 \times \cdots \times V_m = \{(v_1, \dots, v_m) : v_i \in V_i\}$ is a vector space with componentwise operations: $(v_1, \dots, v_m) + (w_1, \dots, w_m) = (v_1 + w_1, \dots, v_m + w_m)$, $\lambda(v_1, \dots, v_m) = (\lambda v_1, \dots, \lambda v_m)$. Zero is $(0, \dots, 0)$.

- Ex. 39, 40: $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$, and $\mathbb{R}^m \times \mathbb{R}^n \cong \mathbb{R}^{m+n}$ by $((x_1, \dots, x_m), (y_1, \dots, y_n)) \mapsto (x_1, \dots, x_m, y_1, \dots, y_n)$.
- Let $V_{(i)} = \{(0, \dots, 0, v, 0, \dots, 0) : v \in V_i\}$ (v in slot i). It is a subspace of the product, isomorphic to V_i.

Proposition (Prop. 35, 36 / LADR 3.92: the product is the direct sum of the $V_{(i)}$)
$V_1 \times \cdots \times V_m = V_{(1)} \oplus \cdots \oplus V_{(m)}$. If the V_i are finite-dimensional, $\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$.

Proof. 1. $(v_1, \dots, v_m) = (v_1, 0, \dots, 0) + \cdots + (0, \dots, 0, v_m)$, and this is the only way to write it as a sum with one term from each $V_{(i)}$, because the term from $V_{(i)}$ must have v_i in slot i.

2. By (W1), the dimension is $\sum \dim V_{(i)} = \sum \dim V_i$.

□

- Concretely: if $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ a basis of W, then $(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)$ is a basis of $V \times W$ (HW4 P5).
- The dimension of a product is the *sum* of the dimensions. (The space with dimension $\dim V \cdot \dim W$ is $\mathcal{L}(V, W)$.)
- The product is also called the *external direct sum*.

Proposition (Prop. 37 / LADR 3.93: products and sums)
For subspaces $U_1, \dots, U_m$ of V, the map $\Gamma\colon U_1 \times \cdots \times U_m \to U_1 + \cdots + U_m$, $\Gamma(u_1, \dots, u_m) = u_1 + \cdots + u_m$, is linear and surjective. It is injective exactly when the sum is direct.

Proof. Surjective by the definition of the sum. $\operatorname{null} \Gamma = \{0\}$ says: $u_1 + \cdots + u_m = 0$ forces all $u_i = 0$. That is (W1). □

- Rank–nullity for Γ:
$$\dim(U_1 + \cdots + U_m) = \sum \dim U_i - \dim \operatorname{null} \Gamma \leq \sum \dim U_i,$$
with equality exactly when the sum is direct. **A sum is direct $\iff$ its dimension is the sum of the dimensions.**

§2.4 Affine subsets and quotient spaces (Ch. 8.2)

The question. Given a subspace U of V, how do we build a vector space in which vectors that differ by an element of U count as the same? Example: in $\mathbb{R}^2$ with $U = \operatorname{span}((0,1))$, ignore the y-coordinate, so $(1,2)$ and $(1,4)$ become equal.

Definition (Def. 22, 23 / LADR 3.95, 3.97)**.** For $v \in V$ and a subspace U, $v + U = \{v + u : u \in U\}$. A subset of this form is an **affine subset**, **parallel** to U. Two affine subsets are parallel if they are parallel to the same subspace.

- Ex. 41: the line $x + y = 1$ in $\mathbb{R}^2$ is $(1,0) + \operatorname{span}((-1,1))$.
- Ex. 42: $\int x^2\,dx = \frac{1}{3}x^3 + C$ is the affine subset $\frac{1}{3}x^3 + \mathbb{R}$ of $\mathcal{P}(\mathbb{R})$, parallel to the constants.
- Prop. 38: an affine subset is parallel to only one subspace. If $v + U = x + W$: write $v = x + w_0$ with $w_0 \in W$. For $u \in U$, $v + u = x + w$ for some $w \in W$, so $u = w - w_0 \in W$. So $U \subseteq W$, and symmetrically $W \subseteq U$.

Proposition (Prop. 39 / LADR 3.101: when two parallel affine subsets are equal)

For a subspace U and $v, w \in V$ the following are equivalent:

$$\text{(i) } v - w \in U \qquad \text{(ii) } v + U = w + U \qquad \text{(iii) } (v + U) \cap (w + U) \neq \varnothing.$$

Proof. 1. (i)⇒(ii): for $u \in U$, $v + u = w + \big((v - w) + u\big) \in w + U$. So $v + U \subseteq w + U$. Since $w - v \in U$ too, the same argument gives $w + U \subseteq v + U$.

2. (ii)⇒(iii): $v = v + 0$ lies in both.

3. (iii)⇒(i): if $v + u_1 = w + u_2$, then $v - w = u_2 - u_1 \in U$.

□

So the affine subsets parallel to U either coincide or do not meet: they cut V into non-overlapping pieces.

Definition (Def. 24 / LADR 3.99, 3.102: quotient space)**.** V/U is the set of all affine subsets of V parallel to U, with

$$(v + U) + (w + U) = (v + w) + U, \qquad \lambda(v + U) = (\lambda v) + U.$$

The zero element is $0 + U = U$.

The idea

An element of V/U is a whole set $v + U$, and the same set has many names: $v + U = v' + U$ whenever $v - v' \in U$. The operations are defined using a name. So there is something to check: if we rename both inputs, does the output set stay the same? If not, the "definition" would give two different answers for one question. This check is called *well-definedness*, and it is needed every time a map out of a quotient is defined by a formula in v.

The check is always the same computation: assume the differences of the names are in U, and show the difference of the outputs is in U. It works because U is closed under addition and scalar multiplication. That is exactly why U must be a subspace.

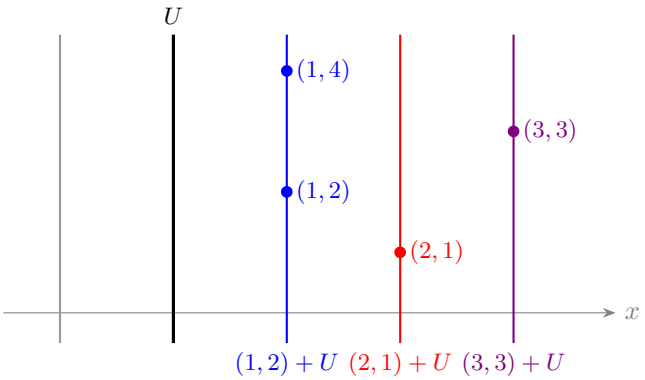
Proof that the operations are well-defined. 1. Suppose $v_1 + U = v_2 + U$ and $w_1 + U = w_2 + U$. By Prop. 39, $v_1 - v_2 \in U$ and $w_1 - w_2 \in U$.

2. $(v_1 + w_1) - (v_2 + w_2) = (v_1 - v_2) + (w_1 - w_2) \in U$, so $(v_1 + w_1) + U = (v_2 + w_2) + U$.

3. $\lambda v_1 - \lambda v_2 = \lambda(v_1 - v_2) \in U$, so $\lambda v_1 + U = \lambda v_2 + U$.

4. The vector space axioms for V/U follow from those of V, applied to the names.

□



$U = \operatorname{span}((0,1))$. Each vertical line is one element of $\mathbb{R}^2/U$.

$(1,2)+U = (1,4)+U$: the same line, since $(1,4)-(1,2) \in U$.

Sum of the blue and red lines: add any point of one to any point of the other. $(1,2)+(2,1) = (3,3)$ and $(1,4)+(2,1) = (3,5)$ both lie on the line $x = 3$.

The quotient $\mathbb{R}^2/U$ for U the y-axis. The answer to "blue line + red line" does not depend on which points are used to compute it.

Shortcuts

- **Isomorphism between spaces of equal finite dimension**: prove injectivity only (solve $Tv = 0$). This also proves existence statements (Ex. 34, 35).
- **T is invertible $\iff$ it sends some (every) basis to a basis** (HW4 P3). So: to build an invertible map, send a basis to a basis.
- **2×2 inverse**: swap the diagonal, negate the off-diagonal, divide by $ad - bc$. Not invertible exactly when $ad - bc = 0$. Over $\mathbb{F}_p$ compute $ad - bc \bmod p$.
- **Square matrices / operators in finite dimension**: $ST = I$ already gives $TS = I$ and $S = T^{-1}$. Check one side.
- $(ST)^{-1} = T^{-1}S^{-1}$: the order reverses.
- **Constructing a map with prescribed behaviour**: extend to a basis and define the map on the basis (left inverse of an injective map, extension of an injective map: R1, HW4 P7).
- $\operatorname{rank}(BC) \le \min(\operatorname{rank} B, \operatorname{rank} C)$ (HW4 P4).
- **Is a sum direct?** Compare $\dim(U_1 + \cdots + U_m)$ with $\sum \dim U_i$.
- **Equality in a quotient**: $v + U = w + U \iff v - w \in U$. Every well-definedness check starts with this line.

Traps

- In infinite dimensions, injective does not give surjective, and $ST = I$ does not give $TS = I$ (derivative and integral).
- A non-square matrix can have a one-sided inverse; it is never invertible (the dimensions differ).
- $\dim(V \times W) = \dim V + \dim W$, not the product.
- $v + U$ is a subspace only when $v \in U$ (then it equals U); otherwise it does not contain 0 (HW4 P6).
- An element of V/U is a set, not a vector of V. The name v in $v + U$ is not determined by the set.
- "Isomorphic" says an isomorphism exists. It does not say which one; different bases give different isomorphisms $V \to \mathbb{F}^n$.

§3 Problem set

HW4 P1. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. (a) Verify by direct multiplication that $A\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right) = (ad-bc)I$. (b) Define $\det A = ad - bc$. Prove that if $\det A \neq 0$ then A is invertible, and derive the formula for A^{-1}. (c) Prove, without any general determinant theory, that if $\det A = 0$ then A is not invertible.

Hint for (c): if A were invertible, multiply the identity in (a) by A^{-1}.

HW4 P2. Decide which are invertible, and find the inverse when it exists: $\left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$ over $\mathbb{C}$, $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right)$ over $\mathbb{F}_5$.

Hint: compute $ad - bc$ in the given field.

HW4 P3 (LADR 3D, Exercise 3). Let V be finite-dimensional and $T\colon V \to V$ linear. Prove that the following are equivalent: (a) T is an isomorphism; (b) for every basis $v_1, \dots, v_n$ of V, the list $Tv_1, \dots, Tv_n$ is a basis of V.

Hint: an injective map sends independent lists to independent lists; then count.

HW4 P4. Suppose BC is defined. (a) Prove $\operatorname{rank}(BC) \leq \min\{\operatorname{rank} B, \operatorname{rank} C\}$. (b) Give an example where the inequality is strict although B and C both have positive rank.

Hint: $\operatorname{range}(BC)$ is contained in $\operatorname{range} B$, and it is the image of $\operatorname{range} C$ under B.

HW4 P5. Let V, W be finite-dimensional. Prove $\dim(V \times W) = \dim V + \dim W$ by constructing a basis of $V \times W$ from bases of V and W.

Hint: $(v_i, 0)$ and $(0, w_j)$.

HW4 P6. Let U be a subspace of V and $v, v' \in V$. (a) Prove that $v + U$ is a subspace of V if and only if $v \in U$; in that case $v + U = U$. (b) Prove that $v + U = v' + U$ if and only if $v - v' \in U$.

Hint: a subspace contains 0.

HW4 P7 (optional). Suppose U is a subspace of a finite-dimensional V and $S\colon U \to V$ is injective. Prove that S extends to an invertible operator $T\colon V \to V$.

Hint: extend a basis of U and, separately, the images of that basis; send the new vectors to the new vectors.

R1 (Practice Midterm 1, Question 2: a left inverse). Let V, W be finite-dimensional and $T\colon V \to W$ injective. (a) For a basis $v_1, \dots, v_n$ of V, prove $Tv_1, \dots, Tv_n$ is independent in W. (b) Prove there is a linear $S\colon W \to V$ with $ST = I_V$.

R2 (Midterm 1 review, Exercise 9). Let U, W be subspaces of a finite-dimensional V with $U \cap W = \{0\}$. Show there is an isomorphism $U \times W \to U \oplus W$.

R3 (Midterm 1 review, Exercise 10). On the space $\mathbb{F}^\infty$ of sequences let $R(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$. (i) Show R is injective but not surjective. (ii) Explain why no operator on a finite-dimensional space has this property.

R4 (Midterm 1 review, Exercise 13). Let $P\colon V \to V$ be linear with $P^2 = P$. Show $V = \operatorname{null} P \oplus \operatorname{range} P$.

R5 (Midterm 1 review, Exercise 12). Let $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ be differentiation. Find bases β of $\mathcal{P}_3(\mathbb{R})$ and γ of $\mathcal{P}_2(\mathbb{R})$ such that

$$\mathcal{M}(D) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$